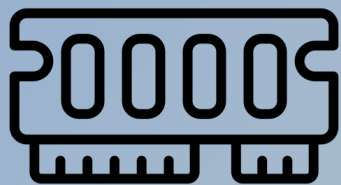


Análisis de RAM de un equipo Windows



Pedro Egea Ortega

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Pregunta 1

Después de comprobar que procesos hay activos en el sistema. Contesta a las siguientes preguntas:

1. ¿Cuántos usuarios han entrado en el sistema? ¿Cuales son sus nombres?

Primer usuario: José

Sesión id: 1

Image file: C:\Users\Pedro\Downloads\DESKTOP-H2VS1BU-20231211-1I

Platform: Windows

Command: windows.pslist.PsList

Command parameters:

☐ Display physical offsets

☐ Process ID

Command Description: Lists the processes present in a particular windows memory image

Run

4016	608	svchost.exe	0xcb8ca1f85080	10	-	1	False	2023-12-11 09:32:39.000000 UTC	N/A	Disabled
544	608	svchost.exe	0xcb8ca9fe52080	8	-	1	False	2023-12-11 09:32:39.000000 UTC	N/A	Disabled
1804	1164	taskhostw.exe	0xcb8ca9fe4d2c0	8	-	1	False	2023-12-11 09:32:39.000000 UTC	N/A	Disabled
3628	608	svchost.exe	0xcb8ca9fae8280	4	-	0	False	2023-12-11 09:32:39.000000 UTC	N/A	Disabled
4212	588	userinit.exe	0xcb8ca9f811340	0	-	1	False	2023-12-11 09:32:39.000000 UTC	2023-12-11 09:33:03.000000 UTC	Disabled
4248	4212	explorer.exe	0xcb8ca9e6340	51	-	1	False	2023-12-11 09:32:39.000000 UTC	N/A	Disabled
4928	608	svchost.exe	0xcb8ca9fd09080	4	-	1	False	2023-12-11 09:32:41.000000 UTC	N/A	Disabled
4576	836	StartMenuExper	0xcb8ca20f2080	9	-	1	False	2023-12-11 09:32:42.000000 UTC	N/A	Disabled
4496	836	RuntimeBroker.	0xcb8ca248f2c0	3	-	1	False	2023-12-11 09:32:42.000000 UTC	N/A	Disabled
4820	836	SearchApp.exe	0xcb8ca2533080	31	-	1	False	2023-12-11 09:32:43.000000 UTC	N/A	Disabled
4952	836	RuntimeBroker.	0xcb8ca24de2c0	5	-	1	False	2023-12-11 09:32:43.000000 UTC	N/A	Disabled
5432	608	svchost.exe	0xcb8ca2650240	3	-	0	False	2023-12-11 09:32:46.000000 UTC	N/A	Disabled
5600	608	svchost.exe	0xcb8ca2651080	3	-	0	False	2023-12-11 09:32:47.000000 UTC	N/A	Disabled
5608	608	svchost.exe	0xcb8ca2653300	2	-	0	False	2023-12-11 09:32:47.000000 UTC	N/A	Disabled
5676	608	svchost.exe	0xcb8ca1aae0c0	4	-	0	False	2023-12-11 09:32:47.000000 UTC	N/A	Disabled
5800	608	svchost.exe	0xcb8ca2515280	1	-	0	False	2023-12-11 09:32:47.000000 UTC	N/A	Disabled

Platform: Windows

Command: windows.envvars.Envvars

Command parameters:

☐ Process ID

☐ Suppress Variables

Refresh Process List

Command Info

Run

Display process environment variables

4248	explorer.exe	0xd61b90	SystemRoot	C:\Windows
4248	explorer.exe	0xd61b90	TEMP	C:\Users\JOS-1\AppData\Local\Temp
4248	explorer.exe	0xd61b90	TMP	C:\Users\JOS-1\AppData\Local\Temp
4248	explorer.exe	0xd61b90	USERDOMAIN	DESKTOP-H2VS1BU
4248	explorer.exe	0xd61b90	USERDOMAIN ROAMINGPROFILE	DESKTOP-H2VS1BU
4248	explorer.exe	0xd61b90	USERNAME	José
4248	explorer.exe	0xd61b90	USERPROFILE	C:\Users\José
4248	explorer.exe	0xd61b90	windir	C:\Windows

Segundo usuario: julian

Sesión id: 2

Command: windows.pslist.PsList													
Command parameters:													
☐ Display physical offsets													
☐ Process ID													
Run													
4444	1164	taskhostw.exe	0xcb8ca20de080	6	-	2	False	2023-12-11 09:44:07.000000 UTC	N/A	Disabled			
3660	6340	userinit.exe	0xcb8ca2494300	0	-	2	False	2023-12-11 09:44:07.000000 UTC	2023-12-11 10:17:53.000000 UTC	Disabled			
7648	3660	explorer.exe	0xcb8ca31680c0	41	-	2	False	2023-12-11 09:44:07.000000 UTC	N/A	Disabled			
8404	8388	msedge.exe	0xcb8ca43020c0	0	-	2	False	2023-12-11 09:44:14.000000 UTC	2023-12-11 09:44:27.000000 UTC	Disabled			
8956	608	svchost.exe	0xcb8ca2fef340	5	-	2	False	2023-12-11 09:44:16.000000 UTC	N/A	Disabled			
5388	836	TextInputHost.exe	0xcb8ca38cc0c0	11	-	2	False	2023-12-11 09:44:19.000000 UTC	N/A	Disabled			
8396	836	StartMenuExper	0xcb8c9fa28080	17	-	2	False	2023-12-11 09:44:23.000000 UTC	N/A	Disabled			

Platform: Windows		Refresh Process List		Display process environment variables	
Command: windows.envvars.Envvars		Command Info			
Command parameters:		Run			
☐ Process ID					
☐ Suppress Variables					
7648	explorer.exe	0x12f1b70	TEMP	C:\Users\julian\AppData\Local\Temp	
7648	explorer.exe	0x12f1b70	TMP	C:\Users\julian\AppData\Local\Temp	
7648	explorer.exe	0x12f1b70	USERDNSDOMAIN	SEVERO.COM	
7648	explorer.exe	0x12f1b70	USERDOMAIN	SEVERO	
7648	explorer.exe	0x12f1b70	USERDOMAIN_ROAMINGPROFILE	SEVERO	
7648	explorer.exe	0x12f1b70	USERNAME	julian	
7648	explorer.exe	0x12f1b70	USERPROFILE	C:\Users\julian	
7648	explorer.exe	0x12f1b70	windir	C:\Windows	

Tercer usuario: antonio

Sesión id: 3

Command:

windows.pslist.PsList

▼

Command Info

Command parameters:

☐ Display physical offsets

☐ Process ID

Run

10280	608	svchost.exe	0xcb8ca188c080	6	-	3	False	2023-12-11	10:22:17.000000	UTC	N/A	Disabled
8892	1164	taskhostw.exe	0xcb8ca454d080	6	-	3	False	2023-12-11	10:22:17.000000	UTC	N/A	Disabled
3496	10956	userinit.exe	0xcb8c9b60a080	0	-	3	False	2023-12-11	10:22:17.000000	UTC	2023-12-11 10:22:48.000000	UTC Disabled
9396	3496	explorer.exe	0xcb8ca38d5300	43	-	3	False	2023-12-11	10:22:17.000000	UTC	N/A	Disabled
4452	5156	msedge.exe	0xcb8ca24d9080	0	-	3	False	2023-12-11	10:22:21.000000	UTC	2023-12-11 10:22:35.000000	UTC Disabled
11796	608	svchost.exe	0xcb8ca59362c0	5	-	3	False	2023-12-11	10:22:25.000000	UTC	N/A	Disabled
12096	836	TextInputHost.	0xcb8ca58ad080	10	-	3	False	2023-12-11	10:22:27.000000	UTC	N/A	Disabled
2892	836	StartMenuExper	0xcb8ca54d3080	20	-	3	False	2023-12-11	10:22:29.000000	UTC	N/A	Disabled
8500	836	RuntimeBroker.	0xcb8ca4ae22c0	6	-	3	False	2023-12-11	10:22:30.000000	UTC	N/A	Disabled
11496	836	SearchApp.exe	0xcb8ca5569080	28	-	3	False	2023-12-11	10:22:31.000000	UTC	N/A	Disabled

Platform: Windows		Refresh Process List		Display process environment variables	
Command: windows.envvars.Envvars		Command Info			
Command parameters:		Run			
☐ Process ID					
☐ Suppress Variables					
9396	explorer.exe	0x871b80	SystemRoot	C:\Windows	
9396	explorer.exe	0x871b80	TEMP	C:\Users\antonio\AppData\Local\Temp	
9396	explorer.exe	0x871b80	TMP	C:\Users\antonio\AppData\Local\Temp	
9396	explorer.exe	0x871b80	USERDNSDOMAIN	SEVERO.COM	
9396	explorer.exe	0x871b80	USERDOMAIN	SEVERO	
9396	explorer.exe	0x871b80	USERDOMAIN ROAMINGPROFILE	SEVERO	
9396	explorer.exe	0x871b80	USERNAME	antonio	
9396	explorer.exe	0x871b80	USERPROFILE	C:\Users\antonio	
9396	explorer.exe	0x871b80	windir	C:\Windows	

Desde kali:

```
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.envvars.Envvars | grep -E "explorer.exe.*USERNAME"
```

```
(pedro@pedro) - [~/ram/volatility3]
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.envvars.Envvars | grep -E "explorer.exe.*USERNAME"
4248ress explorer.exe 0xd61b90 USERNAME José
7648 explorer.exe 0x12f1b70 USERNAME julian
9396 explorer.exe 0x871b80 USERNAME antonio
```

2. ¿Qué navegador ha abierto cada uno de los usuarios?

José abrió Microsoft Edge (msedge.exe)

6864	6604	msedge.exe	0xcb8ca28f4080	0	-	1	False	2023-12-11 09:33:45.000000 UTC	2023-12-11 09:33:50.000000 UTC	Disabled
4936	6864	msedge.exe	0xcb8ca2a6a300	0	-	1	False	2023-12-11 09:33:46.000000 UTC	2023-12-11 09:33:50.000000 UTC	Disabled
6480	6864	msedge.exe	0xcb8ca28e9080	0	-	1	False	2023-12-11 09:33:46.000000 UTC	2023-12-11 09:33:50.000000 UTC	Disabled
1468	6864	msedge.exe	0xcb8ca323a080	0	-	1	False	2023-12-11 09:33:46.000000 UTC	2023-12-11 09:33:50.000000 UTC	Disabled
7540	6864	msedge.exe	0xcb8ca33e70c0	61	-	1	False	2023-12-11 09:33:50.000000 UTC	N/A	Disabled
7564	7540	msedge.exe	0xcb8ca347b080	8	-	1	False	2023-12-11 09:33:50.000000 UTC	N/A	Disabled
7736	7540	msedge.exe	0xcb8ca347d080	15	-	1	False	2023-12-11 09:33:50.000000 UTC	N/A	Disabled
7744	7540	msedge.exe	0xcb8ca347e080	14	-	1	False	2023-12-11 09:33:50.000000 UTC	N/A	Disabled
7772	7540	msedge.exe	0xcb8ca347f080	9	-	1	False	2023-12-11 09:33:50.000000 UTC	N/A	Disabled

julian abrió Firefox (firefox.exe)

8456	836	RuntimeBroker.	0xcb8ca4144080	1	-	2	False	2023-12-11 10:17:47.000000 UTC	N/A	Disabled
7328	7648	SecurityHealth	0xcb8ca39f2080	1	-	2	False	2023-12-11 10:17:52.000000 UTC	N/A	Disabled
7064	9784	firefox.exe	0xcb8ca42ec0c0	54	-	2	False	2023-12-11 10:18:02.000000 UTC	N/A	Disabled
10328	7064	firefox.exe	0xcb8ca3952080	16	-	2	False	2023-12-11 10:18:03.000000 UTC	N/A	Disabled
10368	7064	firefox.exe	0xcb8ca4557080	5	-	2	False	2023-12-11 10:18:03.000000 UTC	N/A	Disabled
10684	7064	firefox.exe	0xcb8ca42f9080	15	-	2	False	2023-12-11 10:18:04.000000 UTC	N/A	Disabled
10776	7064	firefox.exe	0xcb8ca45bc080	5	-	2	False	2023-12-11 10:18:04.000000 UTC	N/A	Disabled
10864	7064	firefox.exe	0xcb8ca45bb0c0	16	-	2	False	2023-12-11 10:18:04.000000 UTC	N/A	Disabled

antonio abrió Google Chrome (chrome.exe)

6484	836	RuntimeBroker.	0xcb8ca58602c0	5	-	3	False	2023-12-11 10:22:45.000000 UTC	N/A	Disabled
10544	9396	SecurityHealth	0xcb8ca41130c0	1	-	3	False	2023-12-11 10:22:50.000000 UTC	N/A	Disabled
13072	9396	chrome.exe	0xcb8ca36da080	35	-	3	False	2023-12-11 10:23:05.000000 UTC	N/A	Disabled
8788	13072	chrome.exe	0xcb8ca58f4080	7	-	3	False	2023-12-11 10:23:05.000000 UTC	N/A	Disabled
11472	13072	chrome.exe	0xcb8ca56d0080	15	-	3	False	2023-12-11 10:23:06.000000 UTC	N/A	Disabled
11328	13072	chrome.exe	0xcb8ca34e70c0	15	-	3	False	2023-12-11 10:23:06.000000 UTC	N/A	Disabled
9656	13072	chrome.exe	0xcb8ca552d080	10	-	3	False	2023-12-11 10:23:06.000000 UTC	N/A	Disabled
14200	836	ShellExperienc	0xcb8ca3125080	10	-	3	False	2023-12-11 10:23:19.000000 UTC	N/A	Disabled
4244	836	RuntimeBroker.	0xcb8ca5fc9300	2	-	3	False	2023-12-11 10:23:20.000000 UTC	N/A	Disabled
12680	4732	OneDrive.exe	0xcb8ca5997080	24	-	3	True	2023-12-11 10:23:34.000000 UTC	N/A	Disabled
13332	836	dllhost.exe	0xcb8ca5996080	9	-	3	False	2023-12-11 10:23:56.000000 UTC	N/A	Disabled

Desde kali:

```
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.pslist.PsList | grep -iE "chrome|firefox|msedge|opera|brave"
```

```
(pedro@pedro) [~/ram/volatility3]
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.pslist.PsList | grep -iE "chrome|firefox|msedge|opera|brave"
6864ress660400.msedge.exe 0xcb8ca28f4080 0 - 1 False 2023-12-11 09:33:45.000000 UTC 2023-12-11 09:33:50.000000 UTC Disabled
4936 6864 msedge.exe 0xcb8ca2a6a300 0 - 1 False 2023-12-11 09:33:46.000000 UTC 2023-12-11 09:33:50.000000 UTC Disabled
6480 6864 msedge.exe 0xcb8ca28e9080 0 - 1 False 2023-12-11 09:33:46.000000 UTC 2023-12-11 09:33:50.000000 UTC Disabled
1468 6864 msedge.exe 0xcb8ca323a080 0 - 1 False 2023-12-11 09:33:46.000000 UTC 2023-12-11 09:33:50.000000 UTC Disabled
7540 6864 msedge.exe 0xcb8ca33e70c0 61 - 1 False 2023-12-11 09:33:50.000000 UTC N/A Disabled
7564 7540 msedge.exe 0xcb8ca347b080 8 - 1 False 2023-12-11 09:33:50.000000 UTC N/A Disabled
7736 7540 msedge.exe 0xcb8ca347d080 15 - 1 False 2023-12-11 09:33:50.000000 UTC N/A Disabled
7744 7540 msedge.exe 0xcb8ca347e080 14 - 1 False 2023-12-11 09:33:50.000000 UTC N/A Disabled
7772 7540 msedge.exe 0xcb8ca347f080 9 - 1 False 2023-12-11 09:33:50.000000 UTC N/A Disabled
6268 7540 msedge.exe 0xcb8ca3613340 19 - 1 False 2023-12-11 09:35:33.000000 UTC N/A Disabled
5476 7540 msedge.exe 0xcb8ca3951340 16 - 1 False 2023-12-11 09:35:33.000000 UTC N/A Disabled
7212 7540 msedge.exe 0xcb8ca320f080 8 - 1 False 2023-12-11 09:35:33.000000 UTC N/A Disabled
2184 7540 msedge.exe 0xcb8ca208d080 15 - 1 False 2023-12-11 09:35:37.000000 UTC N/A Disabled
4804 7540 msedge.exe 0xcb8ca32f7080 10 - 1 False 2023-12-11 09:36:02.000000 UTC N/A Disabled
8404 8388 msedge.exe 0xcb8ca43020c0 0 - 2 False 2023-12-11 09:44:14.000000 UTC 2023-12-11 09:44:27.000000 UTC Disabled
7064 9784 firefox.exe 0xcb8ca42ec0c0 54 - 2 False 2023-12-11 10:18:02.000000 UTC N/A Disabled
10328 7064 firefox.exe 0xcb8ca3952080 16 - 2 False 2023-12-11 10:18:03.000000 UTC N/A Disabled
10368 7064 firefox.exe 0xcb8ca4557080 5 - 2 False 2023-12-11 10:18:03.000000 UTC N/A Disabled
10684 7064 firefox.exe 0xcb8ca42f9080 15 - 2 False 2023-12-11 10:18:04.000000 UTC N/A Disabled
10776 7064 firefox.exe 0xcb8ca45bc080 5 - 2 False 2023-12-11 10:18:04.000000 UTC N/A Disabled
10864 7064 firefox.exe 0xcb8ca45bb0c0 16 - 2 False 2023-12-11 10:18:04.000000 UTC N/A Disabled
11056 7064 firefox.exe 0xcb8ca5c58080 5 - 2 False 2023-12-11 10:18:04.000000 UTC N/A Disabled
11064 7064 firefox.exe 0xcb8ca5c5e080 5 - 2 False 2023-12-11 10:18:04.000000 UTC N/A Disabled
10444 7064 firefox.exe 0xcb8ca5ce7240 16 - 2 False 2023-12-11 10:18:05.000000 UTC N/A Disabled
10472 7064 firefox.exe 0xcb8ca5c9a0c0 20 - 2 False 2023-12-11 10:18:06.000000 UTC N/A Disabled
10628 7064 firefox.exe 0xcb8ca5cbc0c0 12 - 2 False 2023-12-11 10:18:06.000000 UTC N/A Disabled
6372 7064 firefox.exe 0xcb8ca42c7080 12 - 2 False 2023-12-11 10:18:28.000000 UTC N/A Disabled
8860 7064 firefox.exe 0xcb8c9fbba0c0 12 - 2 False 2023-12-11 10:19:21.000000 UTC N/A Disabled
4452 5156 msedge.exe 0xcb8ca24d9080 0 - 3 False 2023-12-11 10:22:21.000000 UTC 2023-12-11 10:22:35.000000 UTC Disabled
13072 9396 chrome.exe 0xcb8ca36da080 35 - 3 False 2023-12-11 10:23:05.000000 UTC N/A Disabled
8788 13072 chrome.exe 0xcb8ca58f4080 7 - 3 False 2023-12-11 10:23:05.000000 UTC N/A Disabled
11472 13072 chrome.exe 0xcb8ca56d0080 15 - 3 False 2023-12-11 10:23:06.000000 UTC N/A Disabled
11328 13072 chrome.exe 0xcb8ca34e70c0 15 - 3 False 2023-12-11 10:23:06.000000 UTC N/A Disabled
9656 13072 chrome.exe 0xcb8ca552d080 10 - 3 False 2023-12-11 10:23:06.000000 UTC N/A Disabled
8484 13072 chrome.exe 0xcb8ca5729080 17 - 3 False 2023-12-11 10:25:05.000000 UTC N/A Disabled
7156 13072 chrome.exe 0xcb8ca6195080 15 - 3 False 2023-12-11 10:25:06.000000 UTC N/A Disabled
1744 13072 chrome.exe 0xcb8ca4ae6080 9 - 3 False 2023-12-11 10:25:08.000000 UTC N/A Disabled
11320 13072 chrome.exe 0xcb8ca58d9340 15 - 3 False 2023-12-11 10:25:34.000000 UTC N/A Disabled
9028 13128 msedgewebview2 0xcb8ca6162080 39 - 3 False 2023-12-11 10:30:27.000000 UTC N/A Disabled
4732 9028 msedgewebview2 0xcb8ca6a2e080 7 - 3 False 2023-12-11 10:30:27.000000 UTC N/A Disabled
12040 9028 msedgewebview2 0xcb8ca598e080 22 - 3 False 2023-12-11 10:30:27.000000 UTC N/A Disabled
3136 9028 msedgewebview2 0xcb8ca598f080 12 - 3 False 2023-12-11 10:30:27.000000 UTC N/A Disabled
10820 9028 msedgewebview2 0xcb8ca5493080 8 - 3 False 2023-12-11 10:30:27.000000 UTC N/A Disabled
12772 9028 msedgewebview2 0xcb8ca61bc240 15 - 3 False 2023-12-11 10:30:28.000000 UTC N/A Disabled
7272 9028 msedgewebview2 0xcb8ca56130c0 15 - 3 False 2023-12-11 10:30:28.000000 UTC N/A Disabled
8684 9028 msedgewebview2 0xcb8ca7a77080 15 - 3 False 2023-12-11 10:30:28.000000 UTC N/A Disabled
7992 9028 msedgewebview2 0xcb8ca56c4080 15 - 3 False 2023-12-11 10:30:28.000000 UTC N/A Disabled
1848 9028 msedgewebview2 0xcb8ca40f6080 15 - 3 False 2023-12-11 10:30:29.000000 UTC N/A Disabled
7676 9028 msedgewebview2 0xcb8ca5949080 15 - 3 False 2023-12-11 10:30:32.000000 UTC N/A Disabled
```

3. ¿Qué programas está ejecutando cada usuario?

José:

```
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.envvars.Envvars | grep -i "USERNAME" | grep -i "José" | sort | uniq
```

```
(pedro@pedro)~[~/ram/volatility3]
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.envvars.Envvars | grep -i "USERNAME" | grep -i "José" | sort | uniq
10116essctfmon.exe 0x2119f801b60canUSERNAMEished José
1184 sihost.exe 0x2f0c44e1b60 USERNAME José
12352 conhost.exe 0x15c6e231b70 USERNAME José
1388 DumpIt.exe 0x1151b70 USERNAME José
15200 msedge.exe 0x21db9403ac0 USERNAME José
15328 smartscreen.ex 0x25d1d8d1b60 USERNAME José
2252 RuntimeBroker. 0x1466ec03380 USERNAME José
4016 svchost.exe 0x1cc9f603380 USERNAME José
4248 explorer.exe 0xd61b90 USERNAME José
4496 RuntimeBroker. 0x1b27b203380 USERNAME José
4576 StartMenuExper 0x28b9c203510 USERNAME José
4804 msedge.exe 0x1af66a03ac0 USERNAME José
4928 svchost.exe 0x26ce8c03380 USERNAME José
4952 RuntimeBroker. 0x264c2803380 USERNAME José
544 svchost.exe 0x1d4f0203380 USERNAME José
5992 Skype.exe 0xc111ee0 USERNAME José
6412 Discord.exe 0xd4c1cb0 USERNAME José
7212 msedge.exe 0x25a54c03ac0 USERNAME José
7540 msedge.exe 0x165e0203ac0 USERNAME José
7744 msedge.exe 0x1d3d3a03ac0 USERNAME José
8100 svchost.exe 0x2a19e603380 USERNAME José
8120 TextInputHost. 0x276f7a034c0 USERNAME José
8168 Skype.exe 0xc0a1ca0 USERNAME José
```

- 4248 explorer.exe (su escritorio)
- 6412 Discord.exe (múltiples instancias)
- 8168 Skype.exe (múltiples instancias)
- 7540 msedge.exe (múltiples instancias) (Microsoft Edge)
- 6596 OneDrive.exe
- 1388 DumpIt.exe (para generar el volcado de la RAM)

julian:

```
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.envvars.Envvars | grep -i "USERNAME" | grep -i "julian" | sort | uniq
```

```
(pedro@pedro)-[~/ram/volatility3]
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.envvars.Envvars | grep -i "USERNAME" | grep -i "julian" | sort | uniq
10284essShellExperienc 0x2789c2034e0canUSERNAMEished julian
10328 firefox.exe 0x1399dce2060 USERNAME julian
13336 ApplicationFra 0x19de9a51b80 USERNAME julian
1732 svchost.exe 0x1a827a03360 USERNAME julian
2312 RuntimeBroker. 0x1f999c03360 USERNAME julian
4160 sihost.exe 0x29623881b40 USERNAME julian
5388 TextInputHost. 0x2486e4034a0 USERNAME julian
5400 svchost.exe 0x20df5203360 USERNAME julian
7064 firefox.exe 0x1fca2941c70 USERNAME julian
7648 explorer.exe 0x12f1b70 USERNAME julian
7844 RuntimeBroker. 0x28ef74033a0 USERNAME julian
8152 svchost.exe 0x1d96ba03360 USERNAME julian
8456 RuntimeBroker. 0x1a37ee03360 USERNAME julian
8956 svchost.exe 0x1dd47803360 USERNAME julian
9076 RuntimeBroker. 0x233d5003360 USERNAME julian
9268 RuntimeBroker. 0x2a051e03360 USERNAME julian
9340 dllhost.exe 0x1f2aaa31b80 USERNAME julian
```

7648 explorer.exe (su escritorio)

7064 firefox.exe (12 instancias)

332 swriter.exe (LibreOffice Writer)

5852 soffice.bin (LibreOffice)

9852 OneDrive.exe

antonio:

```
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.envvars.Envvars | grep -i "USERNAME" | grep -i "antonio" | sort | uniq
```

```
(pedro@pedro)-[~/ram/volatility3]
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.envvars.Envvars | grep -i "USERNAME" | grep -i "antonio" | sort | uniq
10280esssvchost.exe 0x203e4603370canUSERNAMEished antonio
11412 RuntimeBroker. 0x1f296003370 USERNAME antonio
11496 SearchApp.exe 0x1ef706034a0 USERNAME antonio
11796 svchost.exe 0x1d7c1c03370 USERNAME antonio
12096 TextInputHost. 0x26bef0034b0 USERNAME antonio
12312 RuntimeBroker. 0x2cb1a603370 USERNAME antonio
12680 OneDrive.exe 0x5181d20 USERNAME antonio
13072 chrome.exe 0x251ba091c90 USERNAME antonio
13128 zWebview2Agent 0x25bd4231d20 USERNAME antonio
13332 dlhost.exe 0x2d8858a1b90 USERNAME antonio
14132 svchost.exe 0x21c956033b0 USERNAME antonio
14200 ShellExperienc 0x2873f403530 USERNAME antonio
14292 Zoom.exe 0x1c6d4f51c30 USERNAME antonio
15208 RuntimeBroker. 0x25df4e033b0 USERNAME antonio
2604 RuntimeBroker. 0x24eec6033b0 USERNAME antonio
3620 Zoom.exe 0x1d9fcfa1d20 USERNAME antonio
6484 RuntimeBroker. 0x20940003370 USERNAME antonio
7320 smartscreen.ex 0x23c6f531b90 USERNAME antonio
8324 ApplicationFra 0x2e7a84e1b90 USERNAME antonio
8372 svchost.exe 0x1dc2fa03370 USERNAME antonio
8500 RuntimeBroker. 0x291eca03370 USERNAME antonio
8892 taskhostw.exe 0x194bc941b50 USERNAME antonio
9028 msedgewebview2 0x1de61803540 USERNAME antonio
9168 sihost.exe 0x1f9cd0b1b50 USERNAME antonio
9396 explorer.exe 0x871b80 USERNAME antonio
```

9396 explorer.exe (su escritorio)

13072 chrome.exe (8 instancias)

13048 scalc.exe (LibreOffice Calc)

14292 Zoom.exe (2 instancias)

12680 OneDrive.exe

4. ¿A qué hora ha entrado cada usuario en el sistema?

```
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.pslist.PsList | grep "explorer.exe"
```

```
(pedro@pedro)-[~/ram/volatility3]
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.pslist.PsList | grep "explorer.exe"
4248ress421200.0explorer.exe 0xcb8ca1ed6340 51 - 1 False 2023-12-11 09:32:39.000000 UTC N/A Disabled
7648 3660 explorer.exe 0xcb8ca31680c0 41 - 2 False 2023-12-11 09:44:07.000000 UTC N/A Disabled
9396 3496 explorer.exe 0xcb8ca38d5300 43 - 3 False 2023-12-11 10:22:17.000000 UTC N/A Disabled
```

José: 2023-12-11 09:32:39 UTC

julian: 2023-12-11 09:44:07 UTC

antonio: 2023-12-11 10:22:17 UTC

Pregunta 2

Un usuario estaba utilizando el calc de Libreoffice. ¿Qué usuario es y como se llama? ¿Como se llama el archivo abierto? Hay un plugin de volatility que se llama filescan que os puede ayudar.

El usuario que usaba el calc de Libreoffice era antonio y el archivo se llama Temperaturas.ods

```
(pedro@pedro)-[~/ram/volatility3]
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.pslist.PsList | grep -i "calc"
13048ess939600.0scalc.exe 0xcb8ca555f080 1 - 3 False 2023-12-11 10:24:00.000000 UTC N/A Disabled

(pedro@pedro)-[~/ram/volatility3]
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.filescan.FileScan | grep -i "ods"
0xcb8ca4aba6a0.0\Users\antonio\Documents\Temperaturas.ods
```

Pregunta 3

Un usuario estaba utilizando el writer de Libreoffice. ¿Qué usuario es y como se llama? ¿Como se llama el archivo abierto? Hay un plugin de volatility que se llama filescan que os puede ayudar.

El usuario que usaba el writer de Libreoffice era julian y el archivo se llama Volatility.odt

```
(pedro@pedro)-[~/ram/volatility3]
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.pslist.PsList | grep -i "swriter"
332gress764800.0swriter.exe 0xcb8c9be0f080 1 - 2 False 2023-12-11 10:20:43.000000 UTC N/A Disabled

(pedro@pedro)-[~/ram/volatility3]
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.filescan.FileScan | grep -i "odt"
0xcb8ca2665410.0\Users\julian\Documents\volatility.odt
```



```
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.memmap.Memmap --  
pid 14292 --dump
```

[illegible]

Pregunta 5

¿Indica que búsquedas a hecho cada uno de ellos en youtube? ¿Pon la URL de los vídeos que han visto?

Hacemos los dumps de los navegadores:

Edge (José, PID 7540):

```
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.memmap.Memmap --pid 7540 --dump
```

Firefox (julian, PID 7064):

```
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.memmap.Memmap --pid 7064 --dump
```

Chrome (antonio, PID 13072):

```
$ python3 vol.py -f DESKTOP-H2VS1BU-20231211-104241.raw windows.memmap.Memmap --pid 13072 --dump
```

Ahora filtraré para cada usuario con la siguiente expresión regular

```
grep -oE "https://www.youtube.com/(watch?v=[a-zA-Z0-9_-]+|results\?search_query=[^&\" ]+)" | sort | uniq
```

José:

```
$ strings pid.7540.dmp | grep -oE "https://www.youtube.com/(watch?v=[a-zA-Z0-9_-]+|results\?  
search_query=[^&\" ]+)" | sort | uniq
```

julian:

```
$ strings pid.7064.dmp | grep -oE "https://www.youtube.com/(watch?v=[a-zA-Z0-9_-]+|results\?  
search_query=[^&\" ]+)" | sort | uniq
```

antonio:

```
$ strings pid.13072.dmp | grep -oE "https://www.youtube.com/(watch?v=[a-zA-Z0-9_-]+|results\?  
search_query=[^&\" ]+)" | sort | uniq
```



```
(pedro@pedro) ~/ram/volatility3  
$ strings pid.7540.dmp | grep -oE "https://www.youtube.com/(watch?v=[a-zA-Z0-9_-]+|results\?  
search_query=[^&\" ]+)" | sort | uniq  
https://www.youtube.com/results?search_query=vetusta+morla+directo  
https://www.youtube.com/results?search_query=vetusta+morla+directo  
https://www.youtube.com/watch?v=7trU8F8vWGU  
https://www.youtube.com/watch?v=rchYuzGiJoI  
  
(pedro@pedro) ~/ram/volatility3  
$ strings pid.7064.dmp | grep -oE "https://www.youtube.com/(watch?v=[a-zA-Z0-9_-]+|results\?  
search_query=[^&\" ]+)" | sort | uniq  
https://www.youtube.com/results?search_query=vetusta+morla+directo  
https://www.youtube.com/results?search_query=vetusta+morla+directo  
https://www.youtube.com/watch?v=7trU8F8vWGU  
https://www.youtube.com/watch?v=rchYuzGiJoI  
  
(pedro@pedro) ~/ram/volatility3  
$ strings pid.13072.dmp | grep -oE "https://www.youtube.com/(watch?v=[a-zA-Z0-9_-]+|results\?  
search_query=[^&\" ]+)" | sort | uniq  
https://www.youtube.com/results?search_query=los+simpson  
https://www.youtube.com/results?search_query=vetusta+morla+directo  
https://www.youtube.com/results?search_query=vetusta+morla+directo  
https://www.youtube.com/results?search_query={searchTerms}  
https://www.youtube.com/watch?v=7trU8F8vW  
https://www.youtube.com/watch?v=7trU8F8vWGU  
https://www.youtube.com/watch?v=7trU8F8vWGUCCto5  
https://www.youtube.com/watch?v=UcXotVSd1k8  
https://www.youtube.com/watch?v=rchYuzGiJoI
```

Sospechamos que un usuario estaba utilizando el WhatsApp Web ¿Qué usuario es y mira si puedes conseguir alguna información sobre esa sesión? Cuando entramos en el WhatsApp Web aparecen llamadas de este estilo <https://web.whatsapp.com/>

```
(pedro@pedro): ~/ram/volatility[3]
$ strings DESKTOP-HZV518U-20231211-104241.raw | grep -i "whatsapp" | grep -i "appdata" | sort | uniq
kppdata%Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\2181795718meogdaerlo-ts.sqlite-shm
kppdata%Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\2181795718meogdaerlo-ts.sqlite-shm
kppdata%Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\3166453069gcaw.sqlite
kppdata%Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\3166453069gcaw.sqlite-shm
kppdata%Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\ls\data.sqlite-journal
kppdata%Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\2181795718meogdaerlo-ts.sqlite
kppdata%Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\2181795718meogdaerlo-ts.sqlite-shm
kppdata%Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\3166453069gcaw.sqlite
kppdata%Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\3166453069gcaw.sqlite-shm
kppdata%Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\ls\data.sqlite-journal
9xhuntertest+++++>>>>>>successfully started myself+++++>>>>>>senddatatoserverdownloadwhatssapdatabasereadcallog
lcom\hunter/clientenddownWhatsAppDatabasegetInstalledAppsXhunterTestsLackhook
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\3166453069gcaw.sqlite-journal
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\cache\caches.sqlite-shm
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\2181795718meogdaerlo-ts.sqlite
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\2181795718meogdaerlo-ts.sqlite-journal
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\2181795718meogdaerlo-ts.sqlite-wal
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\3166453069gcaw.sqlite
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\idb\3166453069gcaw.sqlite-wal
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\ls\data.sqlite
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\ls\data.sqlite-journal
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https+++web.whatsapp.com\ls\data.sqlite-wal
file:///C:/Users/Julian/AppData/Roaming/Mozilla/Firefox/Profiles/jsk19j6b.default-release/storage/default/https+++web.whatsapp.com/cache/caches.sqlite
file:///C:/Users/Julian/AppData/Roaming/Mozilla/Firefox/Profiles/jsk19j6b.default-release/storage/default/https+++web.whatsapp.com/cache/caches.sqlite-wal
file:///C:/Users/Julian/AppData/Roaming/Mozilla/Firefox/Profiles/jsk19j6b.default-release/storage/default/https+++web.whatsapp.com/ls/data.sqlite
```

Si profundizamos un poco más:

```
$ strings pid.7064.dmp | grep -iE "@s.whatsapp.net" | sort | uniq
```

```

[pedro@pedro] ~/~/ram/volatility3
└─$ strings DESKTOP-HV51BUW-20231211-104241.raw | grep -i 'whatsapp' | grep -i 'julian' | sort | uniq
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\ids\3166453869wacw.sqlite-Journal
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\caches\sqlite-shm
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\ids\218179571fmeegdarlo-ts.sqlite
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\ids\218179571fmeegdarlo-ts.sqlite-Journal
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\ids\218179571fmeegdarlo-ts.sqlite-wal
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\ids\3166453869wacw.sqlite
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\ids\3166453869wacw.sqlite-wal
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\ids\data.sqlite
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\ids\data.sqlite-Journal
C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\ids\data.sqlite-wal
file://C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\caches\sqlite
file://C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\caches\sqlite-shm
file://C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\ids\data.sqlite-wal
file://C:\Users\Julian\AppData\Roaming\Mozilla\Firefox\Profiles\jsk19j6b.default-release\StorageDefault\https==web.whatsapp.com\ids\data.sqlite-wal

[pedro@pedro] ~/~/ram/volatility3
└─$ strings pid.7064.dmp | grep -is '8s.whatsapp.net' | sort | uniq
3789534728s.whatsapp.net
d34657833712s.whatsapp.net

[pedro@pedro] ~/~/ram/volatility3
└─$ strings pid.7064.dmp | grep -is 'whatsapp' | grep -is 'pushname|notify|name' | sort | uniq
24https://web.whatsapp.com/s
C:\Users\artime\2023-12-11T10:20:03.343Z\{"auth":"AWtYf9oyayQxULn0_waUChN29r7QVQVfBHMdsdtkbXNH1Bv8m","hosts":["hostname","media-madi-1.cdn.whatsapp.net","class":"pop","ips":["ip4":"31.13.83.51","ip6":"2a03:2880:f204:c5:f
ace:b0c:0:167"],"fallback":{"hostn

```

Servidor: media-mad1-1.cdn.whatsapp.net (Madrid)